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(54) **DEVICE FOR AUTOMATICALLY
RELEASING CREMATED ASHES**

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(71) Applicant: **Sent Into Space Limited**, Sheffield
(GB)
(72) Inventors: **Alex Baker**, Sheffield (GB); **Chris
Rose**, Sheffield (GB)
(73) Assignee: **Sent Into Space Limited**, Sheffield
(GB)

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Primary Examiner — William L Miller

(74) *Attorney, Agent, or Firm* — The Webb Law Firm

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ABSTRACT

A device for automatically releasing cremated ashes, including a chamber for containing cremated ashes; and a release mechanism configured to selectively release cremated ashes contained in the chamber to the external environment. The release mechanism includes an annular outer member, and an inner member at least partially radially inward of the outer member. The outer member includes an outlet from the chamber. The release mechanism is configured to translate the inner member and the outer member relative to each other along a common longitudinal axis from a first relative position, in which release of cremated ashes in the chamber through the outlet is inhibited, to a second relative position, in which release of cremated ashes in the chamber through the outlet is enabled. In the first relative position, the inner member is in contact with the outer member so as to cover the outlet. In the second relative position, the inner member is spaced from the outer member.

17 Claims, 3 Drawing Sheets

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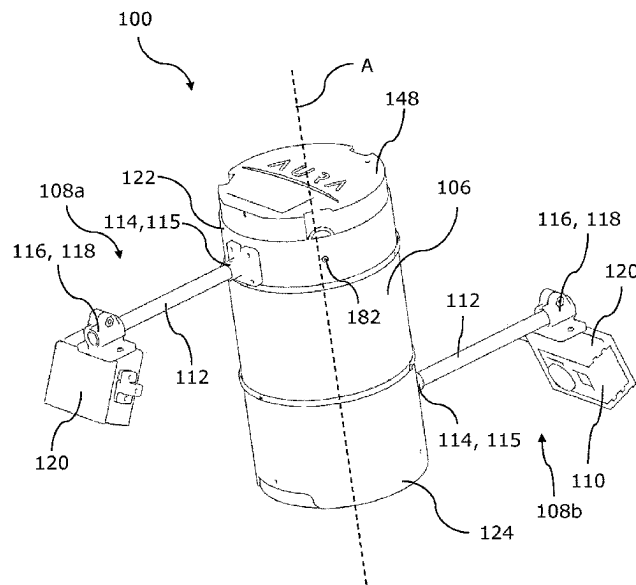
(52) **U.S. Cl.**

CPC **A61G 17/08** (2013.01)

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B64G 1/623; B64G 1/10; B64G 1/2221

See application file for complete search history.



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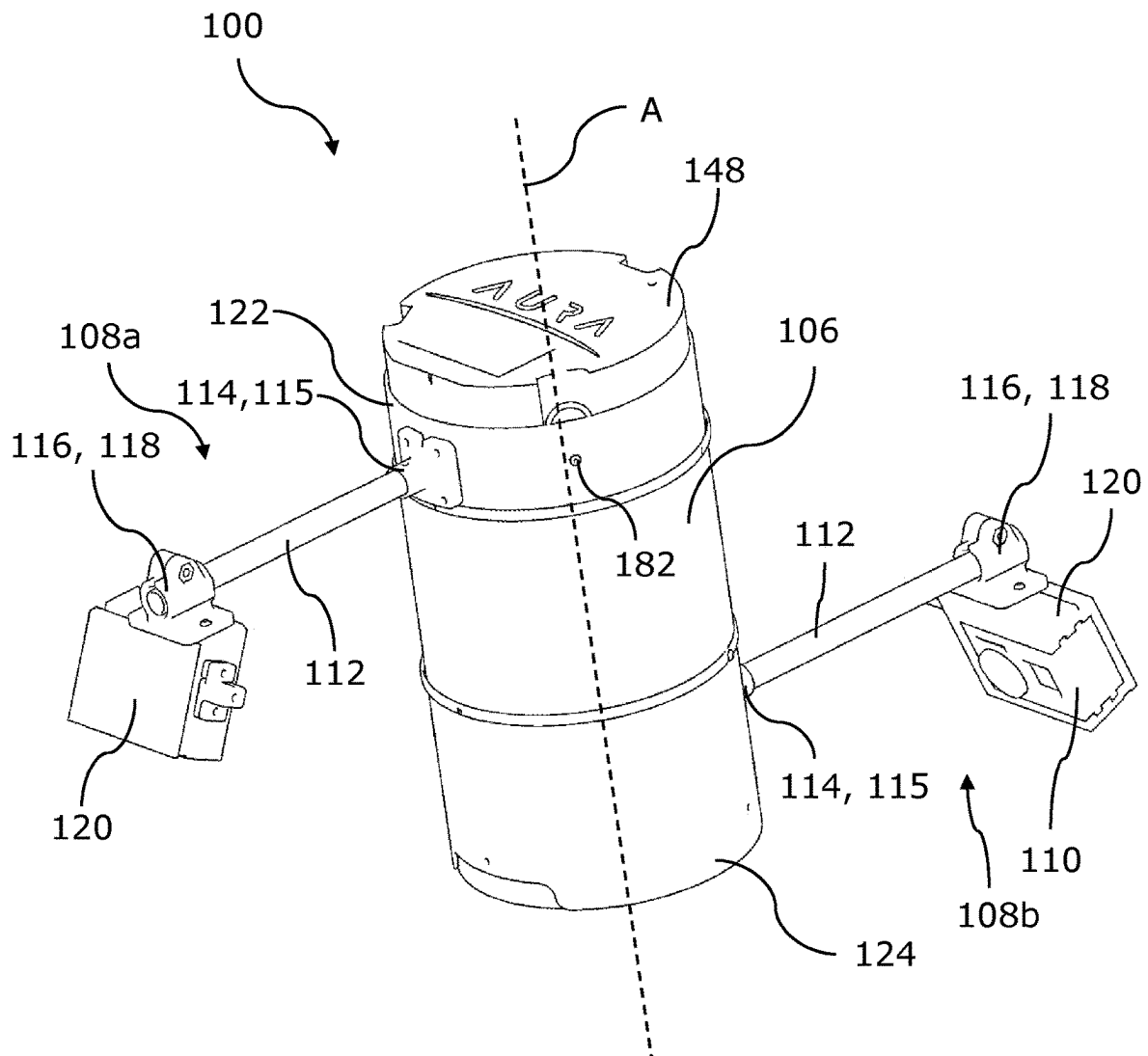


Figure 1

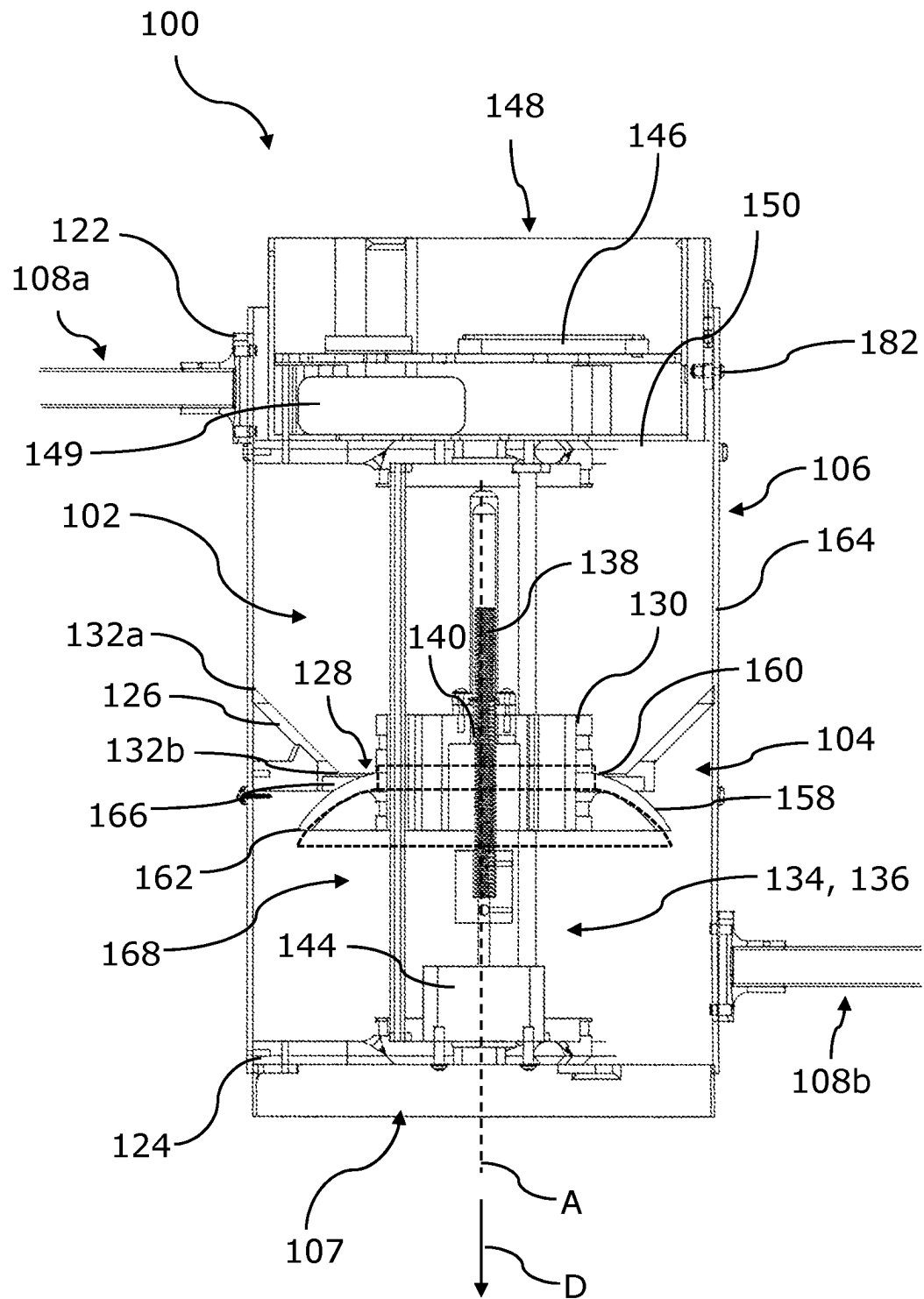


Figure 2

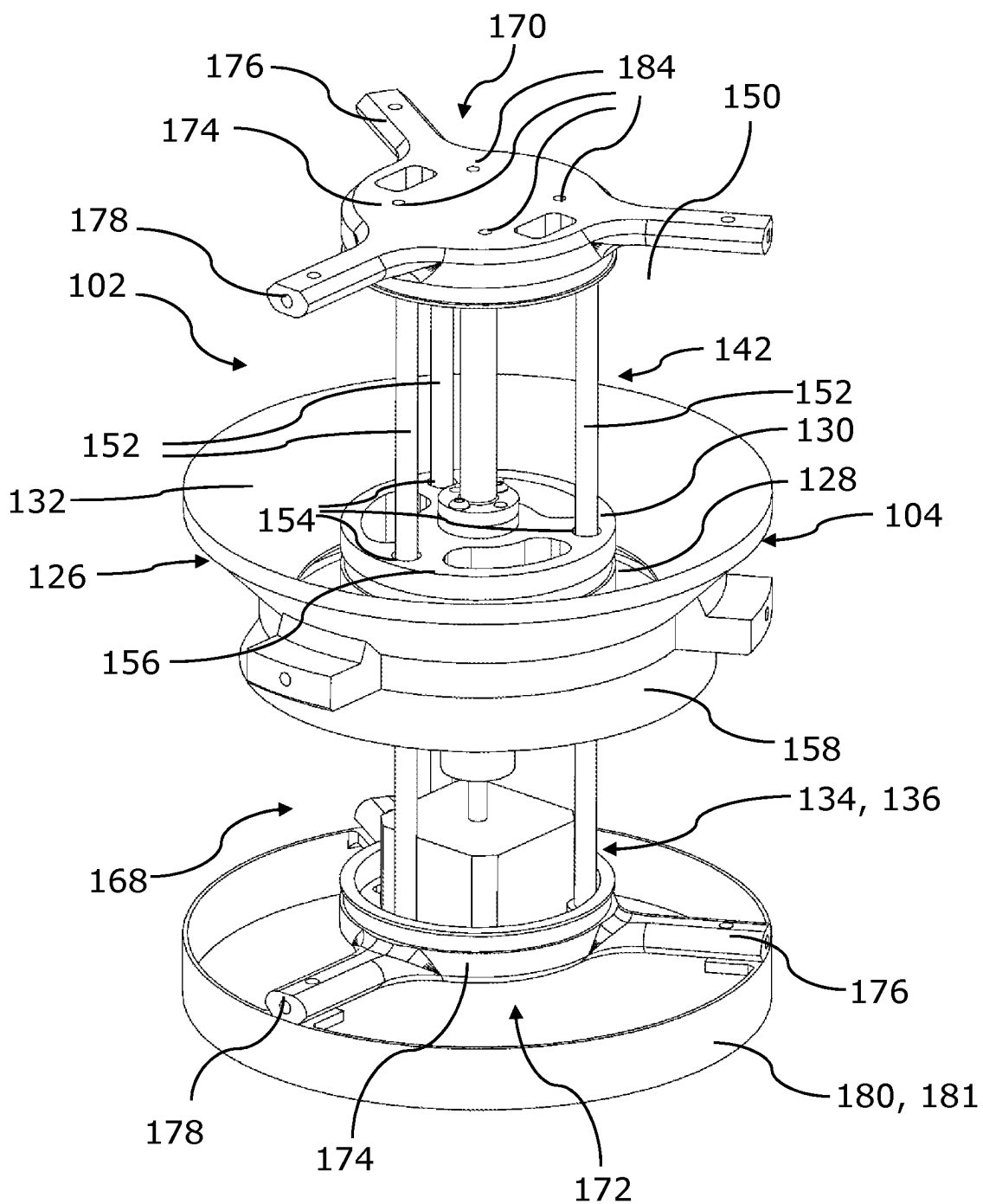


Figure 3

1

**DEVICE FOR AUTOMATICALLY
RELEASING CREMATED ASHES****CROSS-REFERENCE TO RELATED
APPLICATION**

This application claims priority to United Kingdom Patent Application No. 2211401.1 filed Aug. 4, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION**Field of the Invention**

The present teachings relate to a device for automatically releasing cremated ashes, and a method of automatically releasing cremated ashes.

Description of Related Art

In recent years, it has become increasingly popular to scatter the cremated ashes of a deceased person in non-conventional locations, such as in space or high up in Earth's atmosphere.

Devices are known which are configured to elevate the cremated ashes of a deceased person to a high altitude, for example the stratosphere or above, and then release the cremated ashes at this high altitude. To enable relatives and/or friends of the deceased person to watch the cremated ashes being scattered, it is also known to record the ashes as they are released from the device via a camera.

A problem with such known devices is that cremated ashes tend to be released from the device rapidly, resulting in the ash scattering event lasting a relatively short amount of time. The friends and/or relatives of the deceased person commonly wish for a visually appealing and dramatic effect, and may be dissatisfied with the brevity of the ash scattering event.

The present teachings seek to overcome or at least mitigate one or more problems associated with the prior art.

SUMMARY OF THE INVENTION

According to a first aspect of the present teachings, there is provided a device for automatically releasing cremated ashes, comprising: a chamber for containing cremated ashes; and a release mechanism configured to selectively release cremated ashes contained in the chamber to the external environment, the release mechanism comprising an annular outer member, and an inner member at least partially radially inward of the outer member, the outer member comprising an outlet from the chamber. The release mechanism is configured to translate the inner member and the outer member relative to each other along a common longitudinal axis from a first relative position, in which release of cremated ashes in the chamber through the outlet is inhibited, to a second relative position, in which release of cremated ashes in the chamber through the outlet is enabled. In the first relative position, the inner member is in contact with the outer member so as to cover the outlet. In the second relative position, the inner member is spaced from the outer member.

Advantageously, the configuration of the inner member and the outer member helps provide a reliable and robust release mechanism. Moreover, by translating the inner member and the outer member relative to each other along the

2

common longitudinal axis from the first relative position to the second relative position, the rate of release of cremated ashes from the chamber via the outlet can be controlled.

The outer member may comprise an annular sloping surface converging towards the outlet.

Advantageously, the annular sloping surface of the outer member helps maximise the volume of cremated ashes released through the outlet in the second relative position, as well as helping to control the rate of release of cremated ashes from the chamber via the outlet.

The outer member may be at least partially frustoconical.

The annular sloping surface of the outer member may form a maximum angle with the common longitudinal axis in the range of 30 to 60 degrees, preferably in the range of 40 to 50 degrees, for example approximately 45 degrees.

Advantageously, these angles have been found to maximise the volume of ash released through the outlet in second relative position, whilst helping to ensure a controlled release of the ashes through the outlet.

The release mechanism may comprise an actuator arrangement configured to translate the inner member and the outer member relative to each other along the common longitudinal axis from the first relative position to the second relative position.

Advantageously, the actuator arrangement enables automatic controlled release of cremated ashes in the chamber from the device via the outlet.

The actuator arrangement may comprise a linear actuator arranged to engage one of the inner member and the outer member. Actuation of the linear actuator may translate the inner member and the outer member relative to each other along the common longitudinal axis.

Advantageously, the linear actuator provides a reliable and robust means for translating the inner member and the outer member relative to each other from the first relative position to the second relative position.

The linear actuator may be a screw linear actuator.

The linear actuator may comprise a stepper motor.

The device may further comprise a controller configured to control the actuator arrangement.

Advantageously, the controller enables cremated ash to be released from the device automatically without manual input, for example once one or more predetermined conditions have been met. This may enable cremated ash to be released automatically at a desired time, altitude and/or position for example.

The controller may be configured to control the actuator arrangement based on at least one predetermined condition, such as: a predetermined release time; a predetermined altitude of the device; a predetermined global position of the device; and a communication signal received by the controller.

The device may further comprise a controller module releasably mounted to the chamber. The controller may be housed in the controller module.

Advantageously, releasably mounting the controller module to the chamber enables the controller module to be removed from chamber for replacement/inspection/maintenance, as well as improving the modularity of device.

The controller module may provide a closure to an inlet to the chamber when the controller module is mounted to the chamber. The device may be configured such that cremated ashes can be introduced into the chamber via the inlet when the controller module is dismounted from the chamber.

Advantageously, configuring the device such that the controller module provides a closure to the inlet of the

chamber, enables cremated ash to be introduced into and contained within the chamber without requiring a separate dedicated closure.

The inner member may comprise an annular sloping surface arranged to contact the outer member in the first relative position.

Advantageously, the annular sloping surface of the inner member helps maximise the volume of cremated ashes released from the device in the second relative position, as well as helping to control the rate of release of cremated ashes from the device.

The inner member may be at least partially dome shaped.

The outlet may be annular. An outer diameter of the annular sloping surface of the inner member may be greater than a diameter of the outlet.

The annular sloping surface of the inner member may form a maximum angle with the common longitudinal axis in the range of 65 degrees to 85 degrees, preferably 70 degrees to degrees, for example approximately 77 degrees.

Advantageously, these angles have been found to maximise the volume of ash released from the device in the second relative position, whilst helping to ensure a controlled release of the ashes from the device.

A distance between the inner member and the chamber may be greater in the second relative position relative to the first relative position.

Advantageously, configuring the device such that the distance between the inner member and the chamber is greater in the second relative position relative to the first relative position may ensure that cremated ash flows over inner member when released from the chamber, which may help to control release of the cremated ash from the device.

The inner member may comprise an annular sloping surface arranged to contact the outer member in the first relative position. Said annular sloping surface may diverge along the common longitudinal axis from a first end thereof adjacent the chamber to an opposed second end thereof.

Advantageously, such a configuration of the annular sloping surface of the inner member may help to control the release of ash from the device.

The outlet may be annular.

Advantageously, such a configuration of the outlet may help ensure that cremated ash is released from the device as a single stream.

The outer member may be fixed relative to the chamber.

Advantageously, fixing the outer member relative to the chamber may help to maximise the volume in the chamber for contained cremated ashes, which may help to improve the compactness of the device.

The outer member may be mounted to a peripheral wall of the chamber, for example, via one or more fasteners.

The outer member or the inner member may comprise an annular sealing member configured to form a seal with the other of the outer member or the inner member in the first relative position.

Advantageously, the sealing member may help to prevent cremated ash from passing through outlet in the first relative position.

The sealing member may be formed from a flexible polymeric material, such as rubber.

The release mechanism may comprise a guiding arrangement configured to inhibit relative rotational movement between the inner member and the outer member about the common longitudinal axis.

Advantageously, the guiding arrangement may help ensure proper alignment between the inner and outer members from the first relative position to the second relative

position, as well as helping to enable a linear screw actuator to translate the inner member and the outer member relative to each other along the common longitudinal axis.

The guiding arrangement may be configured to constrain relative movement between the inner member and the outer member to along the common longitudinal axis.

The guiding arrangement may comprise one or more elongate guiding members arranged substantially parallel to, and offset from, the common longitudinal axis. One of the inner member and the outer member may comprise one or more complimentary bores, each arranged to receive a respective one of the one or more guiding members.

Advantageously, such a configuration of the guiding arrangement may help ensure that the relative positions of the inner member and the outer member are accurately controlled.

The inner member may comprise the one or more bores. The one or more bores may pass through a central portion of the inner member.

Advantageously, such a configuration helps to improve the compactness of the device.

The central portion may be radially inward of an annular sloping surface of the inner member arranged to contact the outer member in the first relative position.

The device may further comprise a housing. The chamber and the release mechanism may be at least partially housed within the housing.

Advantageously, the housing helps to protect the chamber and the release mechanism from the external environment.

The housing may be substantially cylindrical.

The chamber may be partially defined by the housing.

The device may further comprise one or more camera mounts mounted to the housing. Each camera mount may be for mounting a camera to the device for capturing images of cremated ash released from the device.

Advantageously, the camera mounts enable cameras to be mounted to the device, for capturing images of cremated ash releasing from the device.

The device may further comprise one or more cameras, each camera mounted to one of the one or more camera mounts.

The device may be configured such that at least one of the one or more camera mounts dismounts from the housing when said at least one camera mount impacts a ground surface at a speed in the range of 3 to 5 meters per second.

Advantageously, such a configuration of the camera mounts helps minimise damage to the housing.

The device may comprise a plurality of the camera mounts. The camera mounts may be spaced radially from each other about a release axis of the device. Cremated ashes may be released from the device in a direction substantially along the release axis, in use.

Advantageously, such a configuration of the camera mounts enables different perspectives of the cremated ash releasing from the device to be captured.

The camera mounts may comprise first and second camera mounts radially spaced from each other about the release axis by an angle in the range of 150 to 210 degrees, preferably 170 to 190 degrees, for example approximately 180 degrees.

The device may comprise a plurality of the camera mounts. The camera mounts may be spaced from each other along a release axis of the device. Cremated ashes may be released from the device in a direction substantially along the release axis, in use.

5

Advantageously, such a configuration of the camera mounts enables different perspectives of the cremated ash releasing from the device to be captured.

The camera mounts may comprise first and second camera mounts. The first camera mount may be positioned adjacent a first end of the housing. The second camera mount may be positioned adjacent a substantially opposite second end of the housing.

According to a second aspect of the present teachings there is provided a device for automatically releasing cremated ashes, comprising: a chamber for containing cremated ashes; a release mechanism configured to selectively release cremated ashes contained in the chamber to the external environment; a housing arranged to at least partially house the chamber and the release mechanism; and one or more camera mounts mounted to the housing, each camera mount for mounting a camera to the device for capturing images of cremated ash released from the device. Each camera mount may comprise an elongate member extending away from the housing, a first end of the elongate member mounted to the housing. Each camera mount is configured such that a camera is mountable to an opposite second end of the elongate member.

Advantageously, such a configuration of the camera mounts enables cameras to be mounted to the device at positions spaced from the housing, to enhance the field of view of cremated ashes releasing from the device.

The device may further comprise one or more cameras. Each camera may be mounted to one of the one or more camera mounts.

The device may be configured such that at least one of the one or more camera mounts dismounts from the housing when said at least one camera mount impacts a ground surface at a speed in the range of 3 to 5 meters per second.

The device may comprise a plurality of the camera mounts. The camera mounts may be spaced radially from each other about a release axis of the device. Cremated ashes may be released from the device in a direction substantially along the release axis, in use.

The camera mounts may comprise first and second camera mounts radially spaced from each other about the release axis by an angle in the range of 150 to 210 degrees, preferably 170 to 190 degrees, for example approximately 180 degrees.

The device may comprise a plurality of the camera mounts. The camera mounts may be spaced from each other along a release axis of the device. Cremated ashes may be released from the device in a direction substantially along the release axis, in use.

The camera mounts may comprise first and second camera mounts. The first camera mount may be positioned adjacent a first end of the housing. The second camera mount may be positioned adjacent a substantially opposite second end of the housing.

According to a third aspect of the present teachings, there is provided a modular device for automatically releasing cremated ashes, comprising: a housing module including a chamber for containing cremated ashes; an actuator module; and a controller module housing a controller configured to control the actuator module. The actuator module forms part of a release mechanism at least partially housed within the housing module configured to selectively release cremated ashes contained in the chamber to the external environment. The actuator module is releasably mounted to the housing module. The controller module is releasably mounted to the housing module and/or the actuator module.

6

Advantageously, such a configuration of the device enables rapid replacement/removal of one or more of the housing, the actuator module and the controller module for maintenance and/or inspection.

The release mechanism may be releasably mounted to the housing module.

The controller module may be positioned at or towards a first end of the housing module. The device may be configured to selectively release cremated ashes contained in the chamber to the external environment from an opposed second end of the housing module.

The controller module may provide a closure to an inlet to the chamber when the controller module is mounted to the housing module and/or the actuator module. The device may be configured such that cremated ashes can be introduced into the chamber via the inlet when the controller module is dismounted from the housing module and/or the actuator module.

According to a fourth aspect of the present teachings, there is provided a method of automatically releasing cremated ashes, comprising the steps of:

providing a device comprising a chamber for containing cremated ashes, and a release mechanism configured to selectively release cremated ashes contained in the chamber to the external environment, the release mechanism comprising an annular outer member, and an inner member at least partially radially inward of the outer member, the outer member comprising an outlet from the chamber;

inserting cremated ashes into the chamber; and automatically translating the inner member and the outer member relative to each other along a common longitudinal axis from a first relative position, in which release of the cremated ashes in the chamber through the outlet is inhibited, to a second relative position, in which release of cremated ashes in the chamber through the outlet is enabled.

In the first relative position, the inner member is in contact with the outer member so as to cover the outlet. In the second relative position, the inner member is spaced from the outer member.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments are now disclosed by way of example only with reference to the drawings, in which:

FIG. 1 is an isometric view of a device for automatically releasing cremated ashes according to an embodiment of the present teachings;

FIG. 2 is a cross-sectional view of the device shown in FIG. 1; and

FIG. 3 is an isometric view of internal components of the device shown in FIG. 1.

DESCRIPTION OF THE INVENTION

Referring firstly to FIGS. 1 and 2, there is shown a device 100 for automatically releasing cremated ashes. The device 100 includes a chamber 102 for containing cremated ashes, a release mechanism 104 configured to selectively release cremated ashes contained in the chamber 102 to the external environment, and a housing 106 arranged to at least partially house the chamber 102 and the release mechanism 104.

In the illustrated embodiment, the housing 106 is a substantially cylindrical shell. In alternative embodiments (not shown), the housing 106 has any suitable shape, such as a prism with a polygonal profile for example.

In the illustrated embodiment, the housing **106** is formed from a carbon fibre reinforced polymer. In alternative embodiments (not shown), the housing **106** is formed from any suitable material, such as any suitable plastic or metallic material, for example.

The device **100** includes two camera mounts **108a**, **108b** mounted to the housing **106**. In alternative embodiments (not shown), a single camera mount or more than two camera mounts are mounted to the housing **106**. Each camera mount **108a**, **108b** is suitable for mounting a camera **110** to the device **100** for capturing images of cremated ash released from the device **100**. In the illustrated embodiment, each camera mount **108a**, **108b** mounts one camera **110** to the device **100** (only one camera **110** is visible in the view shown in FIG. 1).

The release mechanism **104** is configured to release cremated ashes contained in the chamber **102** from the device **100** substantially along a release axis A (represented via a dashed line in FIGS. 1 and 2).

With reference to FIGS. 2 and 3, the release mechanism **104** includes an annular outer member **126**. The outer member **126** includes an outlet **128** from the chamber **102**. The release mechanism **104** further includes an inner member **130** at least partially radially inward of the outer member **126**. The inner member **130** and the outer member **126** have a common longitudinal axis A, which in the illustrated embodiment corresponds to the release axis A. In alternative embodiments (not shown), the release axis and the common longitudinal axis have different relative positions.

The release mechanism **104** is configured to translate the inner member **130** and the outer member **126** relative to each other along the common longitudinal A axis from a first relative (closed) position (shown in uninterrupted line in FIG. 2), in which release of cremated ashes in the chamber **102** through the outlet **128** is inhibited, to a second relative (open) position (shown schematically in dashed line in FIG. 2), in which release of cremated ashes in the chamber **102** through the outlet **128** is enabled. In the following, “first relative position” and “closed position”, and “second relative position” and “open position” shall be used interchangeably.

In the closed position, the inner member **130** is in contact with the outer member **126** so as to cover the outlet **128**. In the open position, the inner member **130** is spaced from the outer member **126**. In the open position, the inner member **130** is no longer covering the outlet **128** so as to enable cremated ash contained in the chamber **102** to pass through the outlet **128** and release from the device **100**.

Advantageously, by translating the inner member **130** and the outer member **126** relative to each other along the common longitudinal axis A from the closed position to the open position, the rate of release of cremated ashes from the chamber **102** via the outlet **128** can be controlled. For example, in the open position, the spacing between the inner member **130** and the outer member **126** can be chosen such that the cremated ashes pass through a resulting gap between the outer member **126** and the inner member **130** at a desired rate of release. Moreover, the rate of release of cremated ashes from the device can be changed without changing the position of the release axis A.

In the illustrated embodiment, cremated ashes contained in the chamber **102** are released to the external environment from an opening **107** in the housing **106** at the second end **124** of the housing **106** when the release mechanism **104** is in the open position.

In the illustrated embodiment, the outlet **128** is annular. In alternative embodiments (not shown), the outlet **128** has any suitable shape.

In the illustrated embodiment, the inner member **130** and the outer member **126** are formed from polylactide (PLA) via an additive manufacturing process. In alternative embodiments (not shown), the inner member **130** and the outer member **126** are formed from any suitable material, such as any suitable plastic or metallic material for example, via any suitable process.

In the illustrated embodiment, a distance between the inner member **130** and the chamber **102** is greater in the open position relative to the closed position. Put another way, the inner member **130** moves away from the chamber **102** in a direction D (represented by an arrow in FIG. 2) as the inner member **130** and the outer member **126** are translated relative to each other along the common longitudinal A axis from the closed position to the open position.

In alternative embodiments (not shown), the distance between the inner member **130** and the chamber **102** is less in the open position relative to the closed position. In such embodiments, the inner member **130** may move towards the chamber **102** (upwards in FIG. 2) as the inner member **130** and the outer member **126** are translated relative to each other along the common longitudinal A axis from the closed position to the open position. In such embodiments, the inner member **130** may be wholly located within the chamber **102** in the open position. In such embodiments, the inner member **130** is modified accordingly.

The chamber **102** is partially defined by the housing **106**. In the illustrated embodiment, the chamber **102** is defined by a peripheral wall **164** of the housing **106** and the release mechanism **104**. In alternative embodiments, the chamber **102** is not partially defined by the housing. For example, the chamber **102** may be partially defined by a peripheral wall which is separate to the housing **106**, and which may be at least partially housed within the housing **106**.

The outer member **126** includes an annular sloping surface **132** converging towards the outlet **128**. In the illustrated embodiment, the annular sloping surface **132** converges along the common longitudinal axis A from a first end **132a** of the annular sloping surface **132** furthest from the outlet **128** to a second end **132b** of the annular sloping surface **132** adjacent the outlet **128**.

The chamber **102** is partially defined by the annular sloping surface **132**. In the illustrated embodiment, the annular sloping surface **132** acts as a funnel, which guides cremated ashes in the chamber **102** towards the outlet **128**.

In the illustrated embodiment, the outer member **126** is at least partially frustoconical. In alternative embodiments (not shown), the outer member **126** has any suitable shape.

The annular sloping surface **132** of the outer member **126** forms a maximum angle with the common longitudinal axis A in the range of 30 to 60 degrees, preferably in the range of 40 to 50 degrees. In the illustrated embodiment, the annular sloping surface **132** of the outer member **126** forms a maximum angle with the common longitudinal axis A of approximately 45 degrees. Advantageously, these angles have been found to ensure that substantially all of the cremated ash in the chamber **102** is released through the outlet **128** at a desired controlled rate in the open position.

In the illustrated embodiment, the annular sloping surface **132** has a substantially straight profile as shown in FIG. 2. In alternative embodiments (not shown), the annular sloping surface **132** has any suitable profile.

In alternative embodiments (not shown), the outer member **126** does not include the annular sloping surface **132**,

and, for example, may instead include a surface substantially parallel to or normal to the common longitudinal axis A.

In the illustrated embodiment, the common longitudinal axis A is an axis of rotational symmetry of the annular sloping surface 132 of the outer member 126.

The inner member 130 includes an annular sloping surface 158 arranged to contact the outer member 126 in the closed position. In the illustrated embodiment, the inner member 130 is partially dome shaped. The annular sloping surface 158 has the shape of a truncated dome. In alternative embodiments (not shown), the annular sloping surface 158 is dome-shaped, or has any suitable sloping shape. For example, the annular sloping surface 158 may have a substantially straight profile.

In the illustrated embodiment, cremated ash passing through the outlet 128 in the open position flows over the annular sloping surface 158 of the inner member 130 before exiting the device 100. As such, the inner member 130 impedes the flow of cremated ash from the device 100 in the open position, helping to control the rate of release of cremated ash from the device 100.

In the illustrated embodiment, the common longitudinal axis A is an axis of rotational symmetry of the annular sloping surface 158 of the inner member 130.

In the illustrated embodiment, an outer diameter of the annular sloping surface 158 of the inner member 130 is greater than a diameter of the outlet 128. As such, the annular sloping surface 158 projects radially outward from the outlet 128, for example as shown in FIGS. 2 and 3. Advantageously, this helps ensure that cremated ash flows over a greater area of the inner member 130 when cremated ash is released from the device 100, to help control the release of the cremated ash from the device 100.

The annular sloping surface 158 of the inner member 130 forms a maximum angle with the common longitudinal axis A in the range of 65 degrees to 85 degrees, preferably 70 degrees to 80 degrees. In the illustrated embodiment, the annular sloping surface 158 of the inner member 130 forms a maximum angle with the common longitudinal axis A of approximately 77 degrees. Advantageously, these angles have been found to maximise the volume of ash released from the device 100 in the open position, whilst helping to ensure a controlled release of the ashes from the device 100.

The annular sloping surface 158 of the inner member 130 forms a minimum angle with the common longitudinal axis A in the range of 20 degrees to 40 degrees, preferably 25 degrees to 35 degrees. In the illustrated embodiment, the annular sloping surface 158 of the inner member 130 forms a minimum angle with the common longitudinal axis A of approximately 30 degrees.

In the illustrated embodiment, the annular sloping surface 158 of the inner member 130 diverges along the common longitudinal axis A from a first end 160 of the annular sloping surface 158 adjacent the chamber 102 to an opposed second end 162 of the annular sloping surface 158. In the illustrated embodiment, the outer member 126 is interposed between the chamber 102 and the second end 162 of the annular sloping surface 158.

In alternative embodiments (not shown), in which the distance between the inner member 130 and the chamber 102 is less in the open position relative to the closed position, the annular sloping surface 158 of the inner member 130 converges along the common longitudinal axis A from a first end thereof adjacent the chamber 102 to an opposed second thereof, for example furthest from the chamber 102.

The outer member 126 is fixed relative to the chamber 102. In the illustrated embodiment, the outer member is mounted to the peripheral wall 164 of the chamber 102, for example, via one or more fasteners, such as bolts.

In alternative embodiments (not shown), the inner member 130 is instead fixed relative to the chamber 102.

The outer member 126 includes an annular sealing member 166 configured to form a seal with the inner member 130 in the closed position. The annular sealing member 166 may be formed from a flexible polymeric material, such as rubber for example. In the illustrated embodiment, the annular sealing member 166 is arranged adjacent to and around the outlet 128.

In alternative embodiments (not shown), the inner member 130 instead includes the annular sealing member 166.

The release mechanism 104 includes an actuator arrangement 134 configured to translate the inner member 130 and the outer member 126 relative to each other along the common longitudinal axis A from the closed position to the open position.

In the illustrated embodiment, the actuator arrangement 134 is fixed relative to the outer member 126, and is configured to translate the inner member 130 relative to the outer member 126. In alternative embodiments (not shown), the actuator arrangement 134 is fixed relative to the inner member 130, and is configured to translate the outer member 126 relative to the inner member 130.

In the illustrated embodiment, the actuator arrangement 134 includes a linear actuator 136 arranged to engage the inner member 130. Actuation of the linear actuator 136 translates the inner member 130 and the outer member 126 relative to each other along the common longitudinal axis A.

The linear actuator 136 is fixed relative to the outer member 126. In the illustrated embodiment, the linear actuator 136 and the outer member 126 are both fixed relative to the housing 106. Actuation of the linear actuator 136 translates the inner member 130 relative to the housing 106.

In the illustrated embodiment, the linear actuator 136 is a screw linear actuator configured to rotate a screw 138 about an axis parallel to the common longitudinal axis A. The screw 138 is received in a threaded bore 140 of the inner member 130. The release mechanism 104 includes a guiding arrangement 142, discussed in more detail in the following, configured to inhibit relative rotational movement between the inner member 130 and the outer member 126 about the common longitudinal axis A. As such, rotation of the screw 138 results in the inner member 130 translating relative to the outer member 126 along the common longitudinal axis A from the closed position to the open position.

In the illustrated embodiment, the linear actuator 136 includes a stepper motor 144 configured to rotate the screw 138. Advantageously, the stepper motor 144 enables precise control of the release mechanism 104. In alternative embodiments (not shown), the linear actuator 136 includes any suitable motor.

In alternative embodiments (not shown), the linear actuator 136 does not engage the inner member 130, and instead engages the outer member 126, such that actuation of the linear actuator 136 translates the inner member 130 and the outer member 126 relative to each other along the common longitudinal axis A. In such embodiments, the linear actuator 136 may be fixed relative to the inner member 130 instead of the outer member 126, for example, the inner member 130 and the linear actuator 136 may both be fixed relative to the housing 106. Actuation of the linear actuator 136 may translate the outer member 126 relative to the housing 106.

11

In alternative embodiments (not shown), the actuator arrangement **134** includes any suitable actuator configured to translate the inner member **130** and the outer member **126** relative to each from the closed position to the open position, for example, a hydraulic or pneumatic actuator.

In the illustrated embodiment, the guiding arrangement **142** is configured to constrain relative movement between the inner member **130** and the outer member **126** to along the common longitudinal axis A. The guiding arrangement **142** includes three elongate guiding members **152** arranged substantially parallel to, and offset from, the common longitudinal axis A. In alternative embodiments (not shown), the guiding arrangement **142** instead includes one, two or more than three guiding members **152**.

The guiding members **152** are fixed relative to the outer member **126**.

The inner member **130** includes complimentary bores **154**, each arranged to receive a respective one of the guiding members **152**. The bores **154** and the guiding members **152** are configured such that the inner member **130** can translate relative to the guiding members **152** along the common longitudinal axis A.

In the illustrated embodiment, a linear bearing (not shown) is mounted within each of the bores **154**. Each linear bearing is interposed between the respective bore **154** and the respective guiding member **152**. Advantageously, the linear bearings help to reduce friction between the guiding members **152** and the inner member **130**, and thus help to provide smooth relative translation between the inner member **130** and the guiding arrangement **142**. In the illustrated embodiment, the bores **154** pass through a central portion **156** of the inner member **130**. The central portion **156** is radially inward of the annular sloping surface **158** of the inner member **130**. Advantageously, such a configuration helps to improve the compactness of the device **100**.

In alternative embodiments (not shown), the guiding members **152** are fixed relative to the inner member **130**. In such embodiments, the guiding members **152** may be arranged so as to be received in complimentary bores **154** in the outer member **126**, and such that the outer member **126** can translate relative to the guiding members **152** along the common longitudinal axis A.

In alternative embodiments (not shown), the device **100** includes any suitable guiding arrangement **142**. Alternatively, in embodiments in which the inner member **130** and the outer member **126** are translated relative to each from the closed position to the open position without the screw **138**, the device **100** may not include the guiding arrangement **142**.

The device **100** further includes a controller **146** configured to control the actuator arrangement **134**. In the illustrated embodiment, the controller **146** includes a global positioning device (e.g. GPS), an altimeter, and a communication device for receiving wireless communication signals (e.g. 5G).

The controller **146** may be configured to control the actuator arrangement **134** based on at least one predetermined condition, such as: a predetermined release time; a predetermined altitude of the device **100**; a predetermined global position of the device **100**; and a communication signal received by the controller **146**. For example, the controller **146** may be configured to control the actuator arrangement **134** to translate the inner member **130** and the outer member **126** relative to each other along the common longitudinal axis A from the closed position to the open position based on the at least one predetermined condition.

12

With further reference to FIG. 1, each camera mount **108a**, **108b** includes an elongate member **112** extending away from the housing **106**. Each elongate member **112** includes a first end **114** mounted to the housing **106**. Each elongate member **112** includes a second end **116** opposite to the first end **114**. Each elongate member **112** is configured such that a camera **110** is mountable to the second end **116** of the elongate member **112**.

In the illustrated embodiment, each elongate member **112** is a rod. In alternative embodiments, each elongate member **112** has any suitable shape.

In the illustrated embodiment, each camera mount **108a**, **108b** includes a bracket **118** secured to the second end **116** of the elongate member **112**. The bracket **118** is attached to one of the cameras **110** via a camera casing **120** arranged to at last partially encase the respective camera **110**. In alternative embodiments (not shown), each bracket **118** is directly secured to one of the cameras **110**.

The device **100** is configured such that the camera mounts **108a**, **108b** dismount from the housing **106** when the camera mounts **108a**, **108b** impact a ground surface at a speed in the range of 3 to 5 meters per second.

In the illustrated embodiment, the first end **114** of each elongate member **112** is mounted to the housing **106** via a connector **115** secured to the housing **106**. Each connector **115** includes a crumple zone configured to release the respective elongate member **112** from the connector **115**, and thus from the housing **106**, when the respective camera mount **108a**, **108b** impacts a ground surface at a speed in the range of 3 to 5 meters per second.

Advantageously, such a configuration of the camera mounts **108a**, **108b** helps to minimise damage to the housing **106**, for example when the device **100** is returning to the ground from a high-altitude (e.g. the stratosphere or above), for example via a parachute or other decelerating means.

In alternative embodiments (not shown), the device **100** includes a plurality of cables, each cable connecting one of the camera mounts **108a**, **108b** to the housing **106**. For example, a first end of each cable may be secured to the respective camera casing **120** and a second end of each cable may be secured to the housing **106** via one of the fasteners **182**. Advantageously, the cables help ensure that the camera mounts **108a**, **108b** remain connected to the housing **106** after they are dismounted from the housing **106**, to limit a separation distance between the camera mounts **108a**, **108b** and the housing **106** after the device **100** has impacted a ground surface, for example.

In alternative embodiments (not shown), the device **100** is configured such that the camera mounts **108a**, **108b** dismount from the housing **106** when the camera mounts **108a**, **108b** impact a ground surface at a speed in the range of 3 to 5 meters per second via any suitable means.

In alternative embodiments (not shown), the device **100** is configured such that one or more of the camera mounts **108a**, **108b** do not dismount from the housing **106** when said one or more camera mounts **108a**, **108b** impact a ground surface at a speed in the range of 3 to 5 meters per second. In the illustrated embodiment, the camera mounts **108a**, **108b** are spaced radially from each other about the release axis A. In the illustrated embodiment, the camera mounts **108a**, **108b** include a first camera mount **108a** and a second camera mount **108b**. The first camera mount **108a** and the second camera mount **108b** are spaced radially from each other about the release axis A by an angle in the range of 150 to 210 degrees, preferably 170 to 190 degrees. In the illustrated embodiment, the first camera mount **108a** and the

13

second camera mount **108b** are spaced radially from each other about the release axis A by approximately 180 degrees.

In alternative embodiments (not shown), the camera mounts **108a**, **108b** are not spaced radially from each other about the release axis A.

In the illustrated embodiment, the camera mounts **108a**, **108b** are spaced from each other along the release axis A. In the exemplary embodiment shown in FIGS. 1 and 2, the first camera mount **108a** is positioned adjacent a first end **122** of the housing **106**, and the second camera mount **108b** is positioned adjacent a substantially opposite second end **124** of the housing **106**.

In alternative embodiments (not shown), the camera mounts **108a**, **108b** are not spaced from each other along the release axis A.

The device **100** is modular and includes a plurality of modules, each module releasably mounted to at least one other module. By ‘releasably mounted’, it is intended to mean that each module can be mounted to and dismantled from at least one other module by using only hand tools (e.g. screw driver, allen key, or pliers) or no tools at all.

The plurality of modules includes a controller module **148**, an actuator module **168**, and a housing module **106**. In the illustrated embodiment, the housing module corresponds to the housing **106**.

In the illustrated embodiment, the actuator module **168** includes the actuator arrangement **134**, the inner member **130** and the guiding arrangement **142**.

The actuator module **168** is releasably mounted to the housing module **106**.

With reference to FIG. 3, the actuator module **168** further includes a first mounting member **170** and a second mounting member **172**. The actuator module **168** is releasably mounted to the first end **122** of the housing module via the first mounting member **170**. The actuator module **168** is releasably mounted to the second end **124** of the housing module **106** via the second mounting member **172**.

The first and second mounting members **170**, **172** each include a hub **174** and three legs **176** extending radially outward from the hub **174**. Each leg **176** is mounted to the housing module **106** via a fastener, such as a bolt, passing through the peripheral wall **164** of the housing module **106** and received in a (e.g. threaded) hole **178** in the leg **176**. In alternative embodiments (not shown), each mounting member **170**, **172** includes one, two or more than three legs **176**.

In the illustrated embodiment, the guiding arrangement **142** is mounted to the hubs **174** of the first mounting member **170** and the second mounting member **172**. The actuator arrangement **134** is mounted to the hub **174** of the second mounting member **172**. The legs **176** of the second mounting member **172** are mounted to an annular member **180** receivable within the housing module **106** so as to form a snug fit with an internal surface of the peripheral wall **164**. By ‘snug fit’ it is intended to mean that a peripheral exterior surface **181** of the annular member **180** abuts against the internal surface of the peripheral wall **164** when the annular member **180** is received in the housing module **106**. The annular member **180** helps increase the rigidity of the housing module **106** at the second end **124** thereof.

The controller module **148** houses the controller **146**. The controller module **148** also houses other components such as a battery **149** arranged to power the controller **146** and the actuator arrangement **134**.

The controller module **148** is releasably mounted to the chamber **102**. In the illustrated embodiment, the controller module **148** is releasably mounted to the housing module

14

106 and the actuator module **168**. The controller module **148** is positioned at or towards the first end **122** of the housing module **106**.

The controller module **148** is releasably mounted to the first end **122** of the housing module **106** via a plurality of fasteners **182**, such as bolts, passing through the peripheral wall **164** of the housing module **106** and bores in the controller module **148**. The controller module **148** is releasably mounted to the actuator module **168** via a plurality of fasteners (not shown), such as bolts, passing through the controller module **148** and bores **184** in the hub **174** of the first mounting member **170**.

In alternative embodiments (not shown), the controller module **148** is mounted to only one of the housing module **106** or the actuator module **168**.

The controller module **148** provides a closure to an inlet **150** to the chamber **102** when the controller module **148** is mounted to the chamber **102**. The device **100** is configured such that cremated ashes can be introduced into the chamber **102** via the inlet **150** when the controller module **148** is dismantled from the chamber **102**. In the illustrated embodiment, the inlet **150** is formed between adjacent legs **176** of the first mounting member **170**.

In the illustrated embodiment, the release mechanism **104** is releasably mounted to the housing module **106**. The outer member **126** is releasably mounted to the housing module **106** via one or more fasteners (not shown), such as bolts. As previously described, the inner member **130** and the guiding arrangement **142** form part of the actuator module **168**, which is releasably mounted to the housing module **106**.

In the foregoing description, the controller module **148**, the actuator module **168**, and the housing module **106** are releasably mounted to the corresponding modules via fasteners. In alternative embodiments (not shown), one or more of the controller module **148**, the actuator module **168**, and the housing module **106** are releasably mounted to the corresponding modules via any suitable releasable mounting arrangement, such as a bayonet-type fitting, for example.

In the following, a method of automatically releasing cremated ashes via the device **100** will be outlined.

In a first step, cremated ashes are introduced into the chamber **102**.

Subsequently, the device **100** may be elevated, for example, to the stratosphere or above. The device **100** may be elevated via a self-propelled vehicle, such as an aircraft or a rocket, for example. The aircraft may be a lighter-than-air aircraft such as a weather balloon, for example.

In a subsequent step, the inner member **130** and the outer member **126** are automatically translated relative to each other along the common longitudinal axis A from the closed position to the open position, to enable the cremated ashes contained in the chamber **102** to release from the device **100** via the outlet **128**.

In the foregoing description, the device **100** is configured to automatically release cremated ashes to the external environment. In alternative embodiments (not shown), the device **100** is configured to automatically release any suitable granular material or any suitable liquid contained in the chamber **102** to the external environment via the outlet **128**.

In the foregoing description, the outer member **126** includes the outlet **128**. In alternative embodiments (not shown), the outer member **126**, suitably modified, includes a plurality of outlets **128** from the chamber **102**. In such embodiments, the inner member **130**, suitably modified, may contact the outer member **126** so as to cover the outlets **128** in the closed position. Alternatively, the release mechanism **104** may include a plurality of inner members **130**,

15

each configured to contact the outer member **126** so as to cover one of the outlets **128** in the closed position. The plurality of outlets **128** may be distributed radially about the common longitudinal axis A.

The invention claimed is:

1. A device for automatically releasing cremated ashes, comprising:

a chamber for containing the cremated ashes; and
a release mechanism configured to selectively release the cremated ashes contained in the chamber to the external environment, the release mechanism comprising an annular outer member, and an inner member at least partially radially inward of the outer member, the outer member comprising an outlet from the chamber,

wherein the release mechanism is configured to translate the inner member and the outer member relative to each other along a common longitudinal axis from a first relative position, in which release of the cremated ashes in the chamber through the outlet is inhibited, to a second relative position, in which release of the cremated ashes in the chamber through the outlet is enabled, wherein, in the first relative position, the inner member is in contact with the outer member so as to cover the outlet, wherein, in the second relative position, the inner member is spaced from the outer member, and wherein the inner member comprises an annular sloping surface arranged to contact the outer member in the first relative position.

2. The device of claim 1, wherein the outer member comprises an annular sloping surface converging towards the outlet.

3. The device of claim 2, wherein the annular sloping surface of the outer member forms a maximum angle with the common longitudinal axis in the range of 30 to 60 degrees.

4. The device of claim 1, wherein the release mechanism comprises an actuator arrangement configured to translate the inner member and the outer member relative to each other along the common longitudinal axis from the first relative position to the second relative position.

5. The device of claim 4, wherein the actuator arrangement comprises a linear actuator arranged to engage one of the inner member and the outer member, and wherein actuation of the linear actuator translates the inner member and the outer member relative to each other along the common longitudinal axis.

6. The device of claim 4, further comprising a controller configured to control the actuator arrangement.

16

7. The device of claim 6, further comprising a controller module releasably mounted to the chamber, wherein the controller is housed in the controller module.

8. The device of claim 1, wherein the annular sloping surface of the inner member forms a maximum angle with the common longitudinal axis in the range of 65 degrees to 85 degrees.

9. The device of claim 1, wherein a distance between the inner member and the chamber is greater in the second relative position relative to the first relative position.

10. The device of claim 1, wherein the outer member is fixed relative to the chamber.

11. The device of claim 1, wherein the outer member or the inner member comprises an annular sealing member configured to form a seal with the other of the outer member or the inner member in the first relative position.

12. The device of claim 1, wherein the release mechanism comprises a guiding arrangement configured to inhibit relative rotational movement between the inner member and the outer member about the common longitudinal axis.

13. The device of claim 12, wherein the guiding arrangement comprises one or more elongate guiding members arranged substantially parallel to, and offset from, the common longitudinal axis, and wherein one of the inner member and the outer member comprises one or more complementary bores, each arranged to receive a respective one of the one or more guiding members.

14. The device of claim 13, wherein the inner member comprises the one or more bores, the one or more bores passing through a central portion of the inner member.

15. The device of claim 1, further comprising a housing, wherein the chamber and the release mechanism are at least partially housed within the housing.

16. The device of claim 15, further comprising one or more camera mounts mounted to the housing, each camera mount for mounting a camera to the device for capturing images of the cremated ashes released from the device.

17. A method of automatically releasing cremated ashes, comprising the steps of:

providing the device of claim 1;

inserting the cremated ashes into the chamber; and

automatically translating the inner member and the outer member relative to each other along the common longitudinal axis from the first relative position to the second relative position.

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